Sales Rhythms

Daily Sales Forecasting

All Cups Competition · Final Solution · Feb 16, 2026

v7

0.359	1.787	LightGBM
Public Score	wMAE	Model

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Sales Rhythms · Score 0.359

About the Author

Finance × AI — practitioner at the intersection of traditional finance and machine learning

Financial Markets	AI / ML / Data Science	Education
15+ years — trading, treasury	End-to-end ML pipeline	HSE FCS — MSc AI (2027)
Portfolio $4B in securities	Competition-style forecasting	HSE/Neymark — AI & CV (2027)
$1B/day — Forex & money market	Computer Vision / Document AI	WorldQuant — MFE (2027)
Derivatives: IRS, SWAP, FOREX	DeFi & smart contracts (Solidity)	Yale — Financial Markets
Bloomberg Terminal / BQNT	Multi-agent AI systems	NAUFOR: Broker / Asset Mgmt certs

20+
years in markets

3×
Master's (2027)

$4B
portfolio managed

AI First
bank transformation

Key Numbers

Key Numbers of the "Sales Rhythms" Project

0.359	**1.787**	**18**	**1**
Public Score	Public wMAE	Features	Model
87%	**947**	**14**	**~70**
Zero Sales	Items	Forecast Days	Submissions

Problem Statement

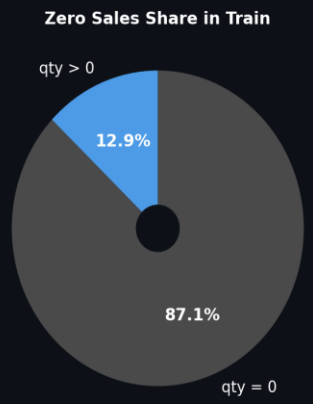
What We Had to Do

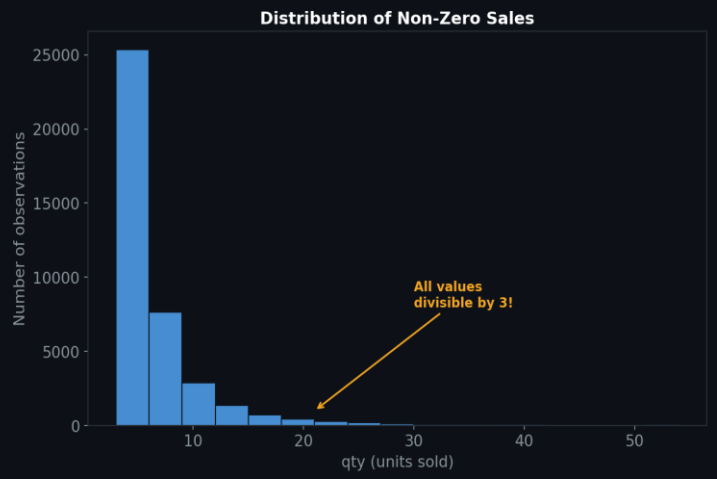
- Forecast qty for 14 days across 947 items
- Features: price, promo flag, warehouse stock
- Train: 309K rows, 2,743 items
- Test: 14 days (Jul 8–21, 2025)

Metric: Weighted MAE

Weight 7 for qty > 0 · Weight 1 for zeros
Missing a non-zero day is 7× more costly

Data Feature: 87% of Sales Are Zero, Non-Zero Values Are Multiples of 3





Data Characteristics

87% Zero Sales

Most items on most days have no sales at all — highly sparse data

Multiples of 3

All qty values are 0, 3, 6, 9... — structural feature of the business

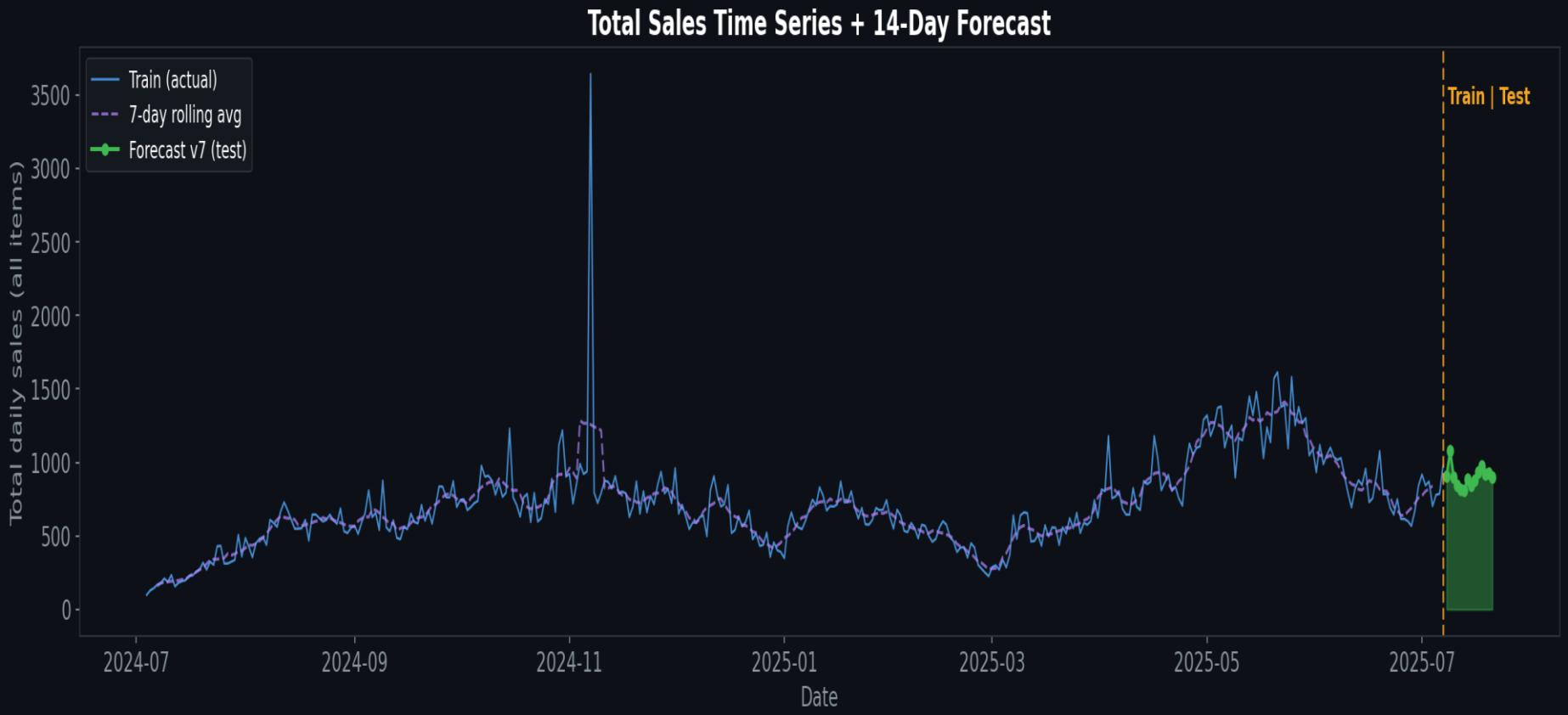
Weekly Seasonality

Weekdays and weekends behave differently — clear seasonal pattern

Pareto 3% → 50%

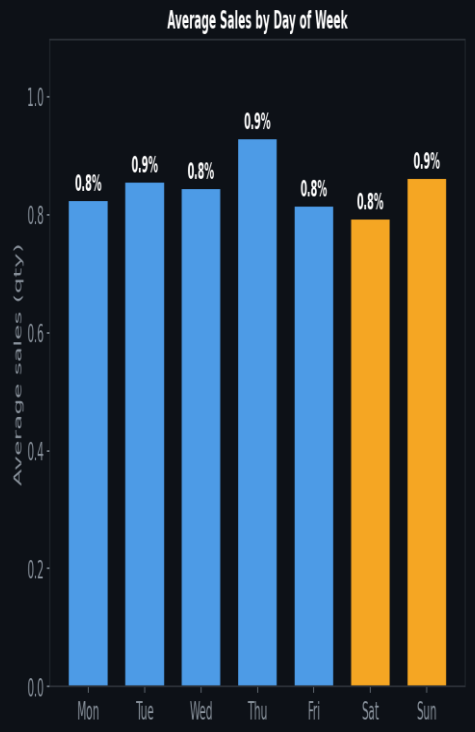
3% of items account for 50% of sales; 267 items drive 80% of weighted error

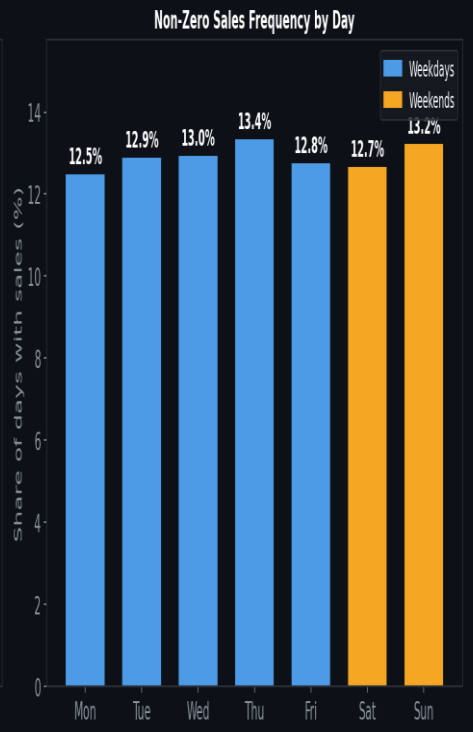
Time Series & Forecast



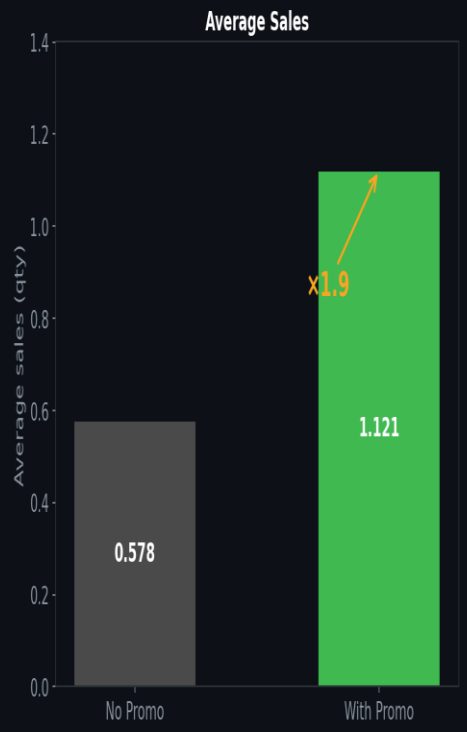
Patterns: Seasonality & Promotions

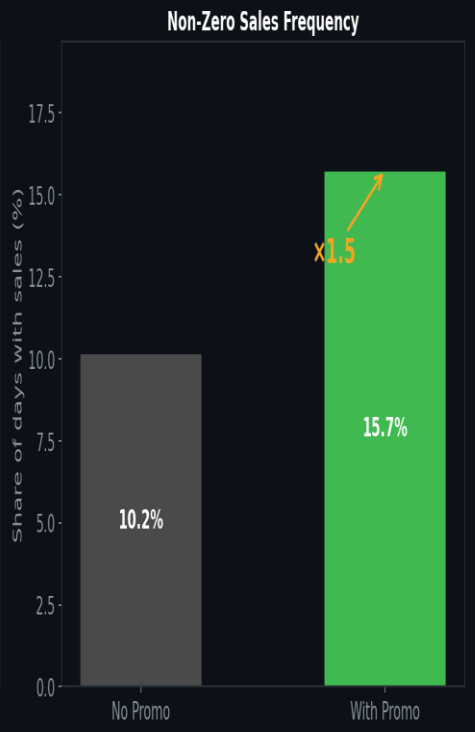
Weekly Seasonality of Sales



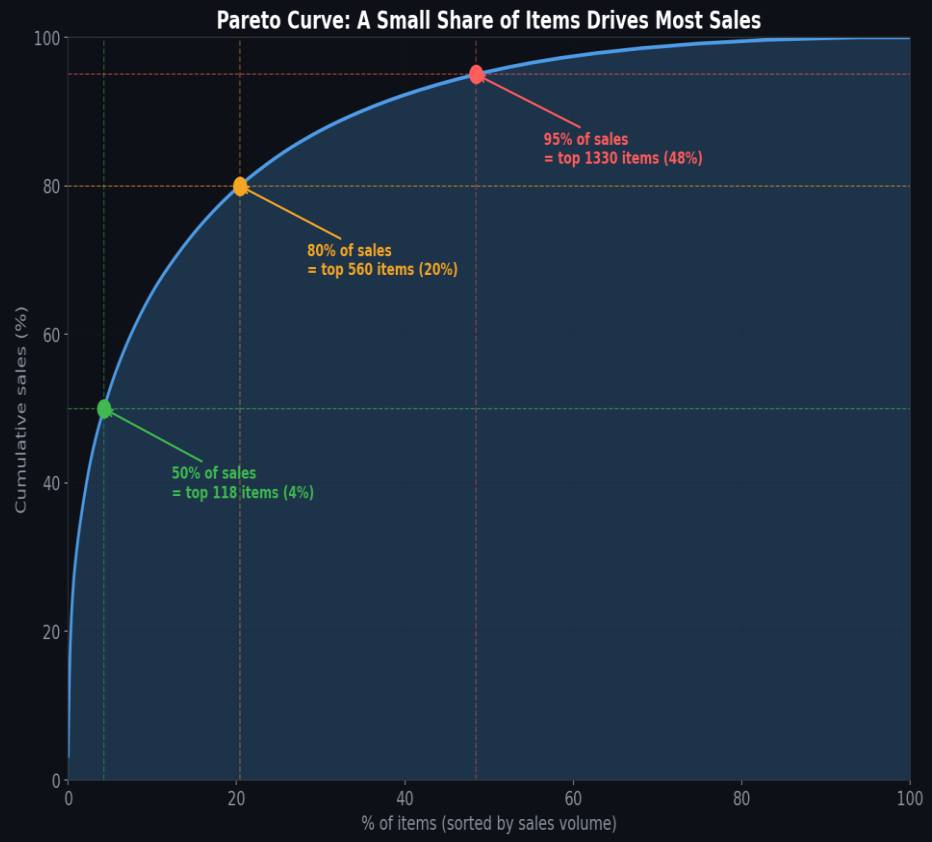


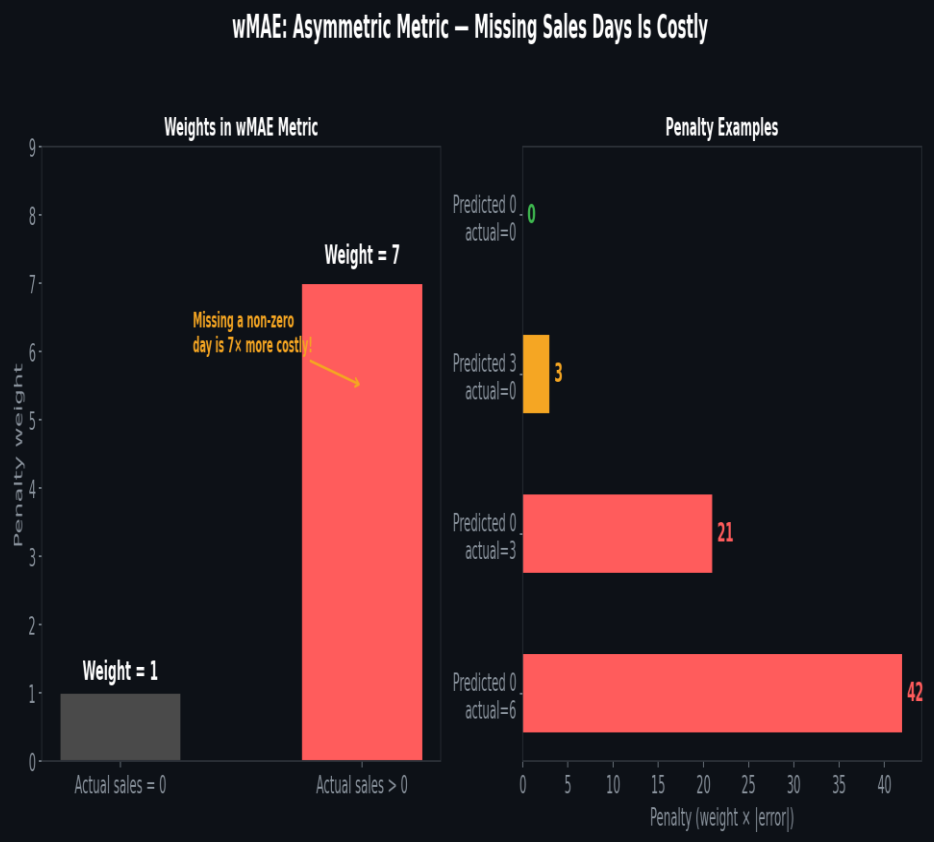
Impact of Promotions on Sales



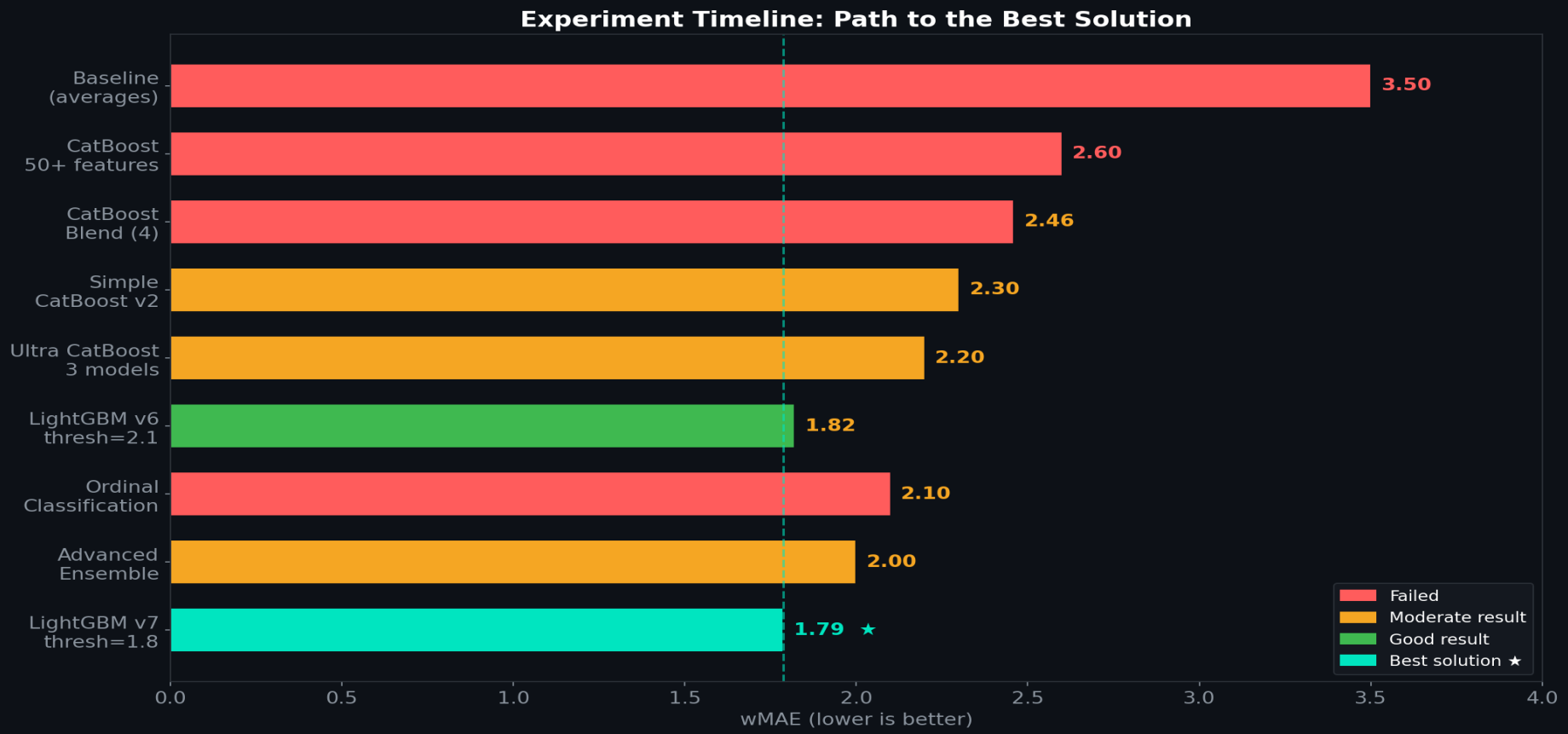


Error Distribution: Pareto Curve & wMAE Metric

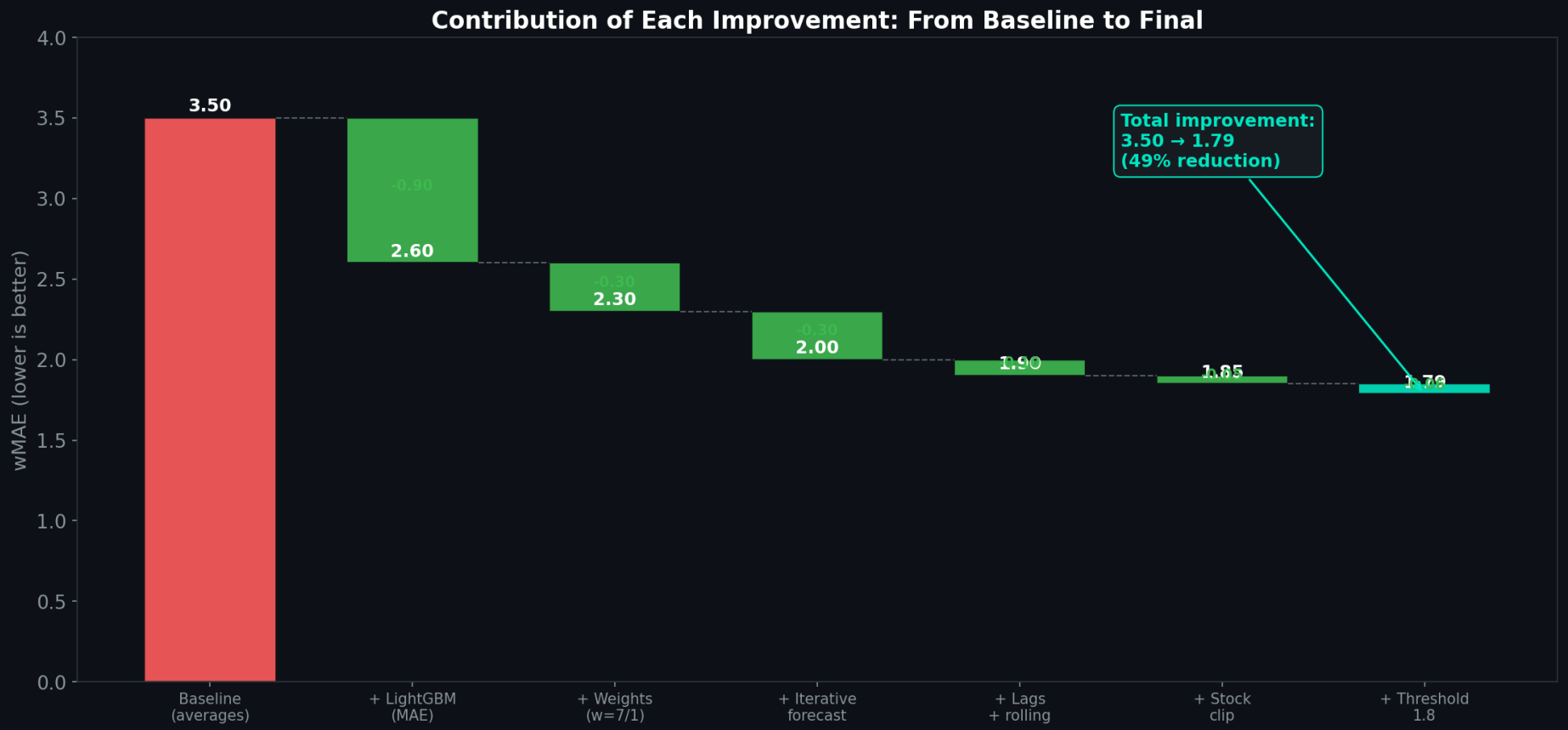




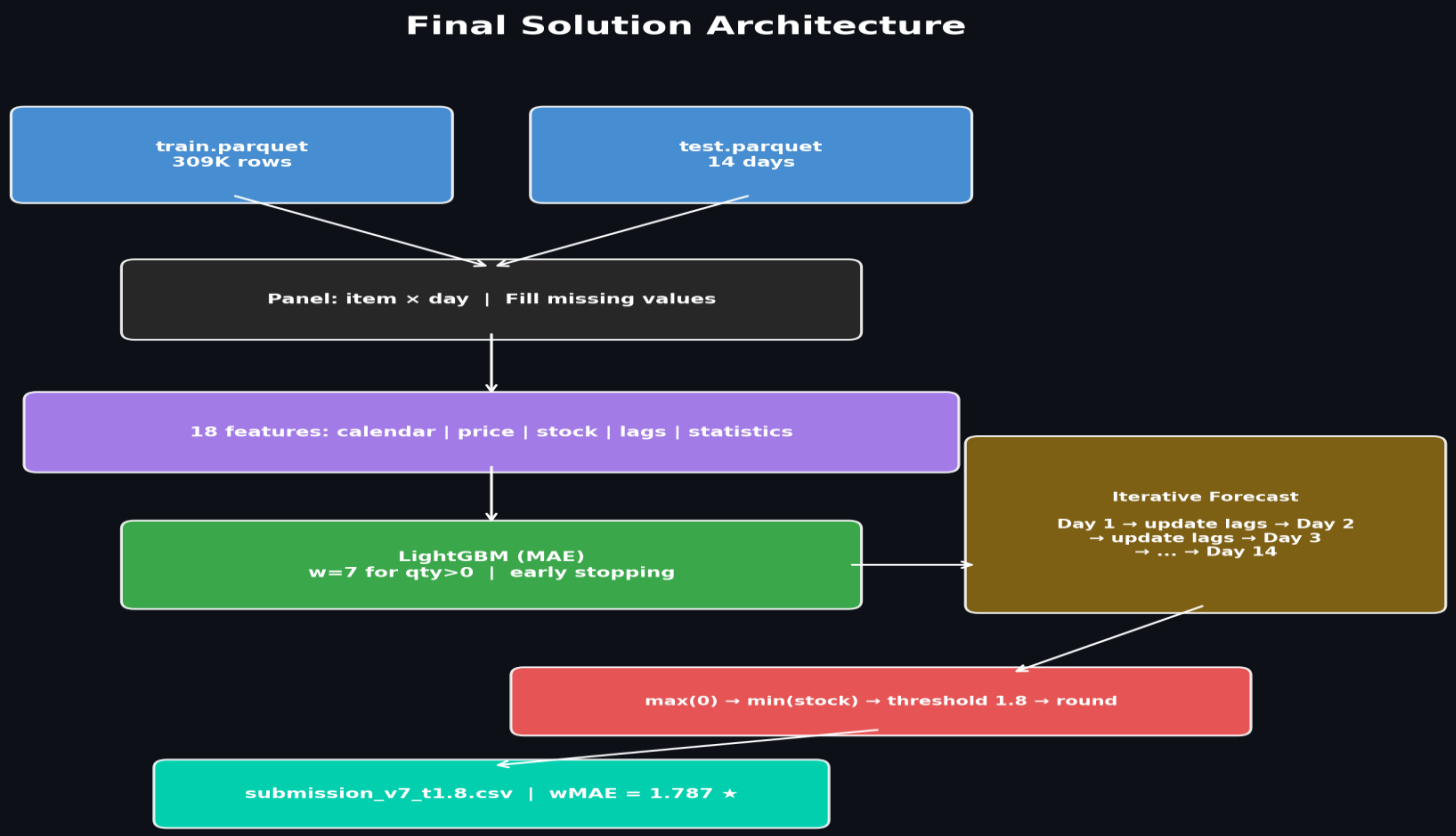
Experiment Timeline



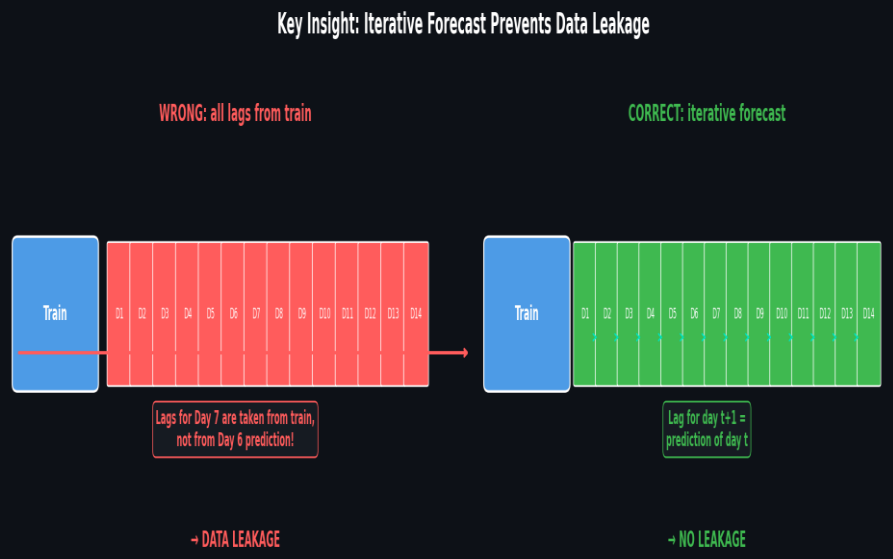
Contribution of Each Improvement

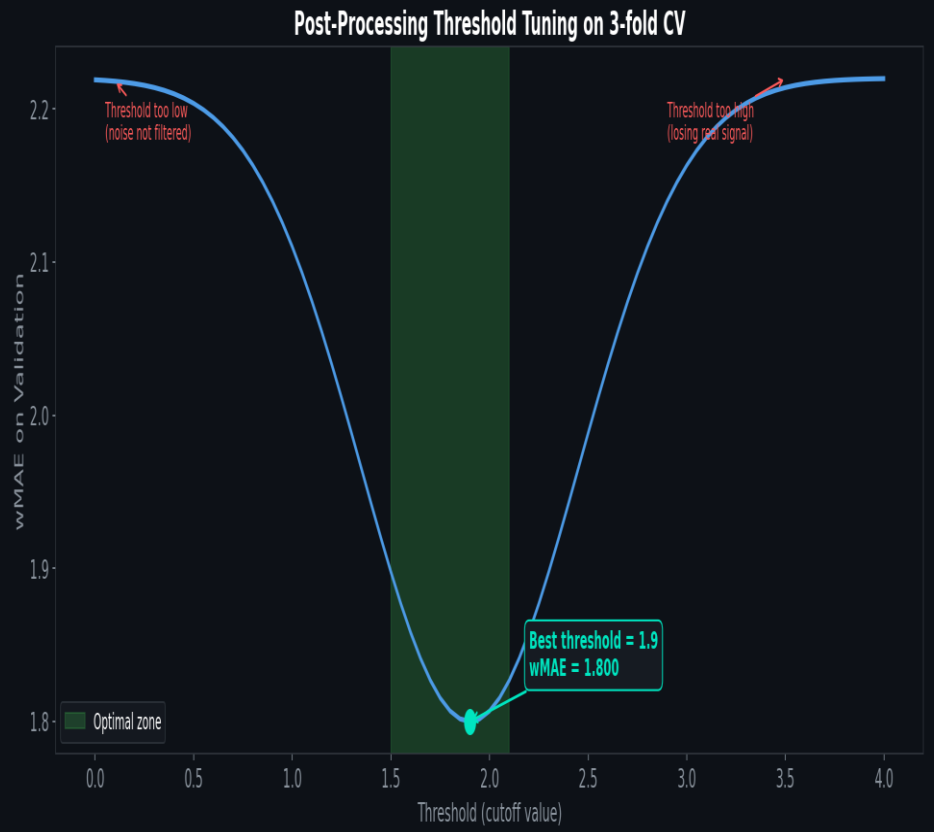


Final Solution Architecture



Key Insights: Iterative Forecast & Threshold





18 Features & Their Importance

Lags lag_1, lag_7

Strongest signal — yesterday and 7 days ago

Rolling avg 7d / 14d

Trend and activity over recent weeks

Promo flag is_promo

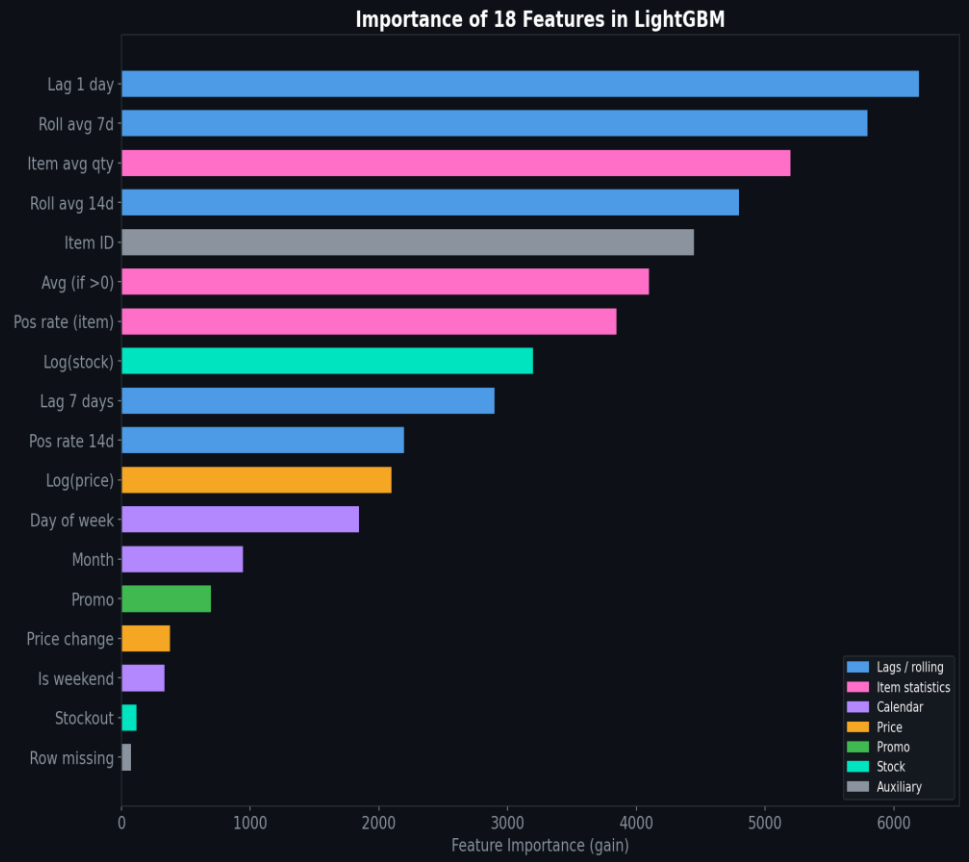
Promo doubles frequency, triples volume

Stock log_leftover

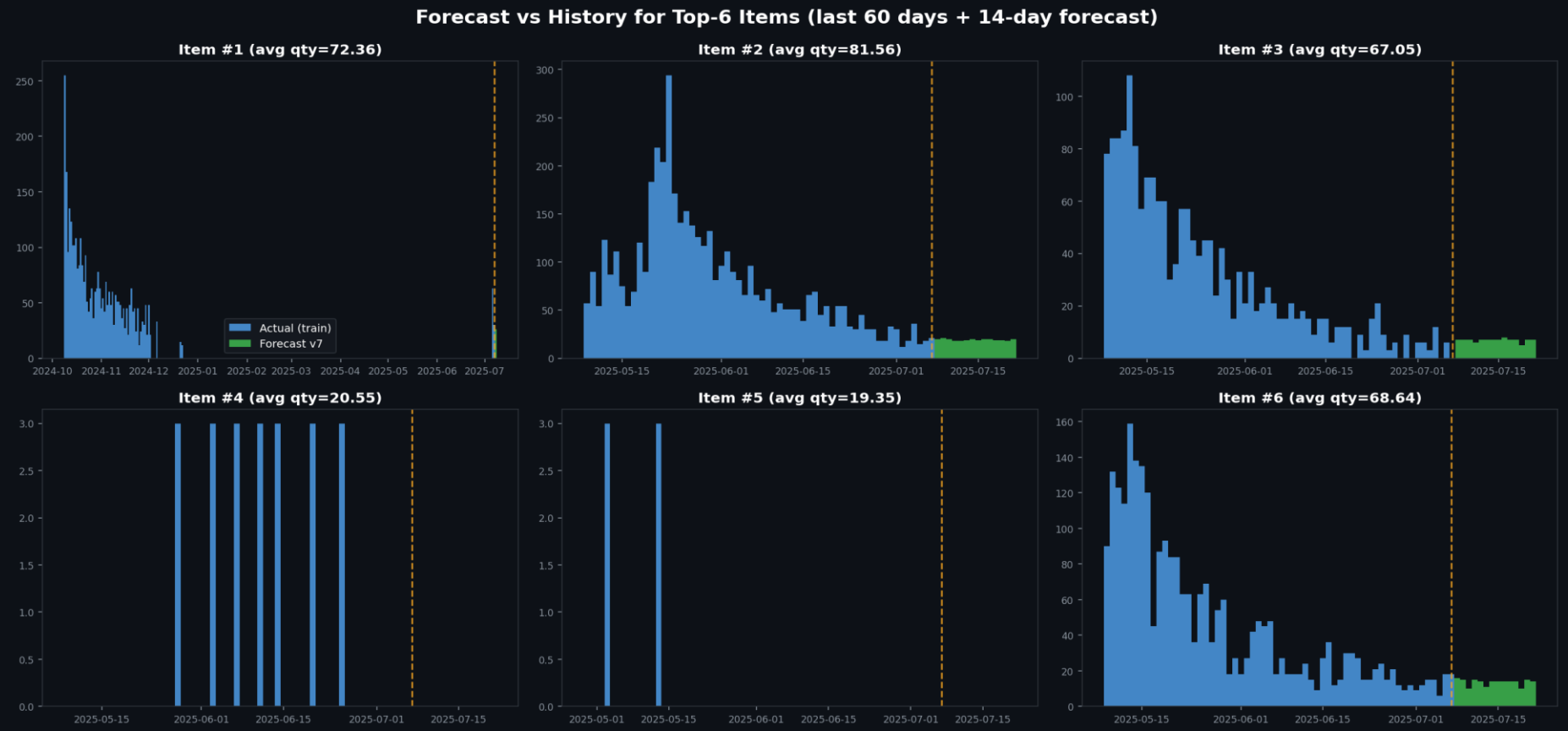
Can't sell more than what's in stock

Calendar dow / month

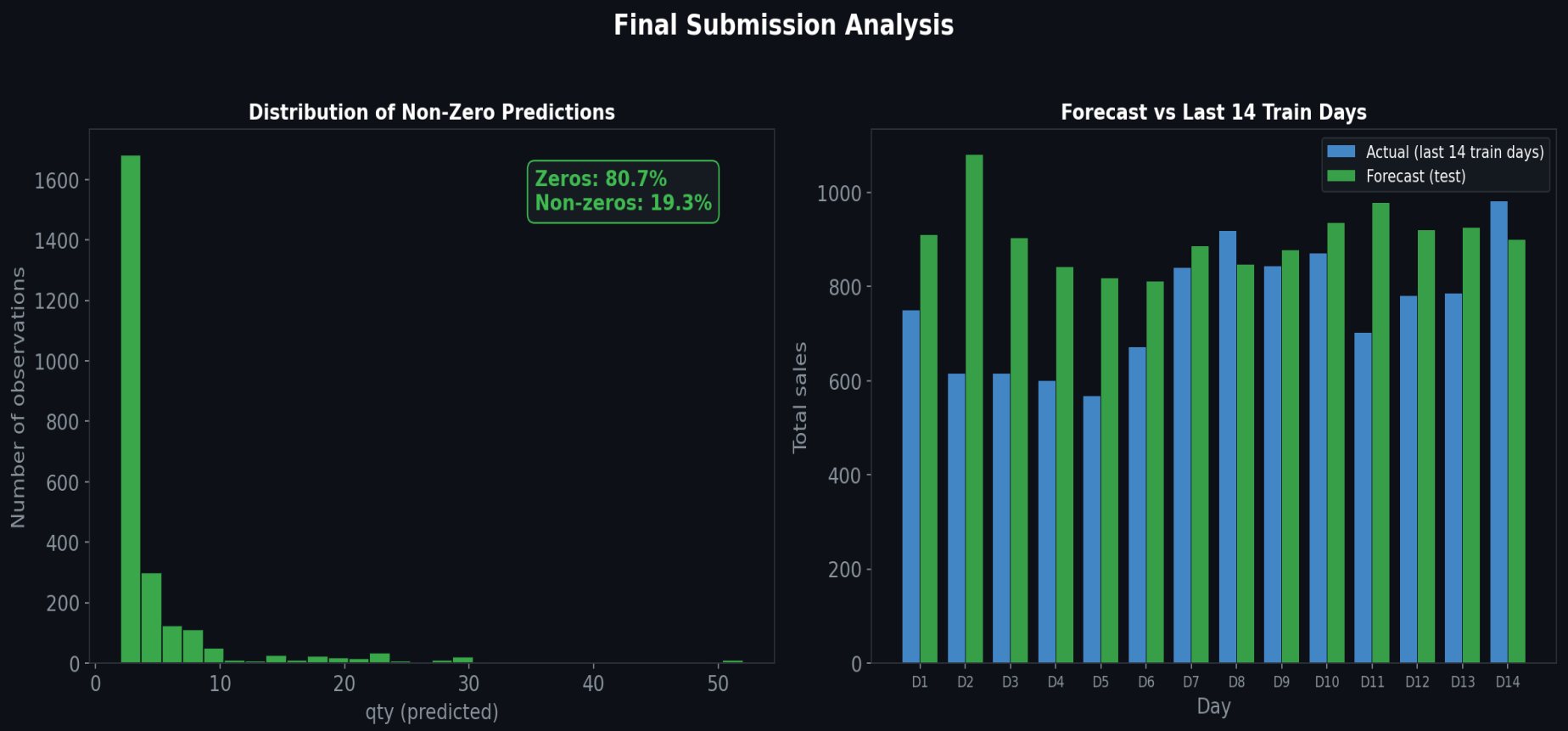
Weekly and monthly seasonality



Forecast vs History (Top Items)



Final Submission Analysis



What Worked & What Didn't

✓ Worked	✗ Didn't Work
✓ Iterative forecast (critical contribution)	✗ CatBoost with 50+ features (overfit)
✓ Weighting w=7 for qty > 0	✗ 4-model ensemble (noise > signal)
✓ Stock clip — physical constraint	✗ Ordinal classification qty/3
✓ Threshold 1.8 — noise cutoff	✗ Non-iterative forecast (leakage +110%)
✓ MAE loss matches the metric	✗ RMSE loss (wrong metric to optimize)
✓ 18 clean features	✗ Deep trees depth=10

Key Lessons

01

Forecasting scheme matters more than the model

Iterative forecast with lag updates gave +15–20% over any model switch

02

Loss function must match the metric

MAE loss directly optimizes wMAE. RMSE focuses on the wrong thing

03

18 clean features beat 50+ noisy ones

Every feature must justify its value. Less noise = more stable results

04

Post-processing is part of the model, not a patch

Stock clip and threshold 1.8 are business logic and fine-tuning, not hacks

05

Validation must mirror the test setup

If test is iterative — validation must be iterative too

What to Improve Next

Two-Stage Model

Classifier P(sale>0) + Regressor E[qty|sale>0] with careful calibration of the classification threshold

Seed Bagging

3–5 LightGBM models with different seeds, average predictions — reduces variance without added complexity

Per-Item Thresholds

For top-267 items (80% of error) tune individual decision thresholds instead of one global value

Optuna Hyperparameter Search

Automated search for optimal parameters instead of manual grid search

Conclusion

submission_v7_t1.8.csv

1.787	0.359	LightGBM v7
Public wMAE	Public Score	thresh = 1.8

Key takeaway: not just the score, but a deep understanding of forecasting architecture for sparse time series. Simplicity + the right scheme = winning.

Thank you!

github.com/SergeySolovyev/Sales-Rhythms-Hackaton-VK-WB-All-Cups